

AMENDMENT NO. 5 NOVEMBER 2022

TO

IS 649 : 1997 METHODS OF TESTING STEEL SHEETS FOR MAGNETIC CIRCUITS OF POWER ELECTRICAL APPARATUS

(Second Revision)

(Page 21, clause 49, sub heading) — Insert sub heading after the heading of the clause:

‘Method 1’.

[Page 21, clause 49, line 3 (see also Amendment No. 3)] — Substitute the following for the sentence:

‘It is intended for non-oriented electrical sheet or strip covered under IS 648 and Grain oriented electrical sheet and strip covered under IS 3024’ for ‘It is intended for grain oriented and non-oriented electrical sheet or strip.’

(Page 21, clause 50, line 8) — Substitute ‘measure of the ductility’ for ‘measure of the brittleness’.

(Page 22, clause 51.3, line 3) — Insert the following at the end of the clause:

‘Test is carried out under controlled conditions shall be made at a temperature of 27 ± 5 °C.’

(Page 22, clause 52.4) — Insert the following at the end of the clause:

‘In case of grain oriented sample along the long axis that is, in the direction of rolling only to be considered.’

(Page 22, clause 53.1) — Substitute following for the existing Fig. 9:

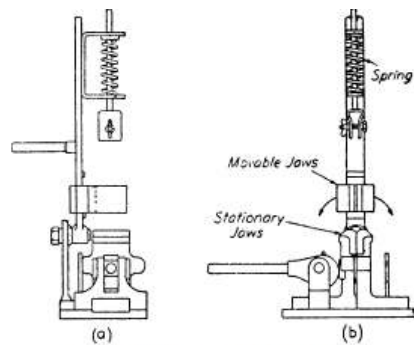


FIG. 9 APPARATUS FOR BEND TEST

(Page 22, clause 53.2) — Substitute following for the existing FIG.10

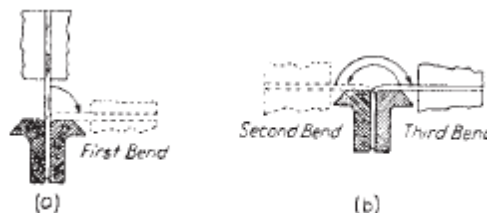


FIG.10 DIAGRAM ILLUSTRATING METHOD OF MAKING BENDS AND ARROWS INDICATE THE COUNT – STATING AS FIRST BEND, SECOND BEND ETC

Price Group 1

Amendment No to IS 649 : 1997

(Page 22, clause **53.2**, line 6) — Substitute ‘bend’ for ‘bent’.

(Page 22, clause **53.3**) — Substitute following for the existing clause:

‘**53.3** Bend at a uniform rate without shock. If necessary, reduce the rate of bending to ensure that heat generated does not affect the result of the test.’

(Page 22, clause **53.4**) — Insert new clause

‘**53.4** Do not interrupt the testing between successive bends if the fracture/crack or failure has not occurred.’

(Page 22, clause **53.5**) — Insert new clause

‘**53.5** First 90° bend is counted as ‘First Bend’. Thereafter the test piece shall be bent through 180° alternatively in opposite directions. One bend consists of bending the test piece through 180°. The following bend shall be made in the opposite direction as shown in Fig. 10. For instance, when sample is bent through 180° in opposite direction, after first bend, is counted as ‘Second Bend’ and again when it is bent through 180° in opposite direction for returning to its position after first bend is counted as ‘Third Bend’ (as shown in Fig 10). The testing between successive bends shall not be interrupted. The test shall be continued until the fracture/crack appears.’

(Page 22, clause **55**) — Insert the following at the end of the clause:

‘It can be used as an alternative method to Method 1, in case full width evaluation is required and agreed between manufacturer and supplier.’

[Page 22, clause **55**, Annex A, **A-1**, last line (see also Amendment No. 2)] — Substitute ‘measure of the ductility’ for ‘measure of the brittleness’

[Page 22, clause **55**, Annex A, **A-4.1**, last line (see also Amendment No. 2)] — Insert following at the end of the clause:

‘Test is carried out under controlled conditions at a temperature of 27 ± 5 °C.’

(Page 23, clause **61.3**) — Substitute the following for the existing clause:

‘When evaluating steel strip narrower than 610 mm, method 1 can be used to determine the ductility.’

(MTD 04)

Publication Unit, BIS, New Delhi, India

TABLE 1: DETAILS ON GAZETTE NOTIFICATION OF Amendment No. 5 to IS 649 : 1997

Dept .	No. &, Year of the Amendment & Title of the Indian Standards	Date of Establish-ment*	Synopsis (Yes/No)	Whether Product std.(Yes/ No)	If Product Standard		Justification					Date of Implement ation
					No. of Licenses	Proposed period for implementa tion	Health	Safety	Protection	Effici ency	Economic impact	
							High/ Low					
1	2	3	4	5	6A	6B	7A	7B	7C	7D	7E	8
MTD	IS 649:1997 Methods for testing steel sheets for magnetic circuits of power electrical apparatus (second revision) Amendment - 5 Doc : MTD 04 (13187)	Date of Approval of CA	N	N	NA	*	NA	Low	Low	Low	Low	*

*To be proposed by CMD-2

**Draft Format to be enclosed with each document, being published by
Publication Department**

Sl. No.	Query about the Document	Answer of the Query
1.	Document Number and Title	MTD/04/ IS 649 :1997 Methods for testing steel sheets for magnetic circuits of power electrical apparatus Amendment - 5
2.	Division Council/Chairperson	Shri Raman Srivastava
3.	Sectional Committee Chairperson Member Secretary	Shri Nirvik Banerjee /Shri Arun Pucchakayala
4.	Date of publication of original standard/past revision & current revision, (in case of revision)	NA
5.	Whether the document is for publishing Indigenous standard or Adopted standard	Indigenous
6.	Need for the Revision (in case of revision)	NA
7.	The substance and the advantages or impact of the new standard or revision	NA
8.	Details of current or prospective users of the standard (Academia/Industry/Institutions/Certification)	Industry, Certification and Academia
9.	The concerned Ministry/Regulatory Body, if applicable	DPIIT, Ministry of Steel
10.	The publicity or dissemination plan for potential users	Through BIS home website and BIS social network websites
11.	Any other relevant information about the document	BIS approved labs for testing electrical steel may be sensitized.